

4001CMD: Mathematical Skills for Computing Professionals

Week 1.2: Logic I: Propositions and Proofs

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1.2 Propositions and Proofs

- 2 Thinking with numbers
- 3 Propositions and Proofs
 - Terminology
 - Compound Statements
 - Existential statements
 - Universal Statements
 - Proof by Contradiction
 - Summary on Proofs



Example (1)

Statement. 15 is a multiple of 3. Or $\exists k \in \mathbb{N} : 15 = 3k$

Let's unpack:

- ▶ the statement "15 is multiple of 3" mathematically can mean (identically)
 - If we divide 15 by 3, then there is zero remainder

$$15 \div 3 = k + 0$$

- Or we can find some (whole) number k , which is when multiplied by 3, will produce 15:

$$3 \times k = 15$$

- The same as saying:

$$\exists k \in \mathbb{N} : 15 = 3k$$

 $\exists$

– "there exist",
that"

 $k \in \mathbb{N}$

– k is in the set of Natural numbers

 $:$

– "such

Example

Statement. 20 is a multiple of 3. $\exists k \in \mathbb{N} : 20 = 3k$

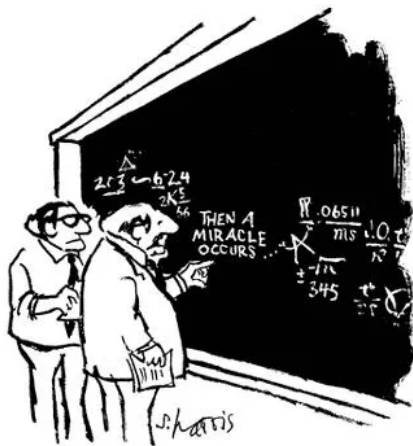
Proof.

If $20 = 3k$ for some positive integer k , then

- ▶ since $18 = 3(6)$ and $21 = 3(7)$,
- ▶ this integer k would need to lie between 6 and 7.
- ▶ There are no such integers.
- ▶ Hence, **this proposition is false.**



- ▶ A **proposition** is a statement that is either true or false.
- ▶ An **axiom**, **postulate**, or **assumption** is a statement that is taken to be true, to serve as a premise or starting point for further reasoning and arguments.
- ▶ In classic philosophy, an axiom is a statement that is so evident or well-established, that it is *accepted without controversy or question*.
- ▶ A true statement is called a **theorem**.
- ▶ A false statement is a **nontheorem**. Its negation is a theorem.
- ▶ A mathematically and logically valid explanation for why a theorem is true is a **proof**.



"I think you should be more explicit here in step two."

A **proof** is a logical argument that tries to show that a statement is true. In math, and computer science, a proof has to be well thought out and tested before being accepted. But even then, a proof can be discovered to have been wrong.

Definition (Proof)

A **proof** is a finite sequence of statements, each of which is either an **axiom**, an **assumption**, or follows from the preceding statements by a rule of **inference**, concluding with the statement that is being proven.

In English, we can modify, combine, and relate propositions with words such as “not” ($\neg$), “and” ($\wedge$), “or” ($\vee$), “implies” ($\implies$), and “if-then”. For example, are the following statements true?

Example (2)

200 is a multiple of 2 **and** 5.

Example (3)

81 is a multiple of 3 **and** 5.

Example (4)

81 is a multiple of 3 **or** 5.

Example (5)

60 is a multiple of 6 **or** 10.

Truth Tables:

Negation (NOT):

P	$\neg P$
T	F
F	T

AND and OR:

P	Q	$P \wedge Q$	$P \vee Q$
T	F	F	T
F	T	F	T
T	T	T	T
F	F	F	F

Compound propositions: implication if ... then ($\rightarrow$ or $\implies$)

The least intuitive connecting word is “**implies**.”

Definition (implication)

An implication is true exactly when the if-part is false or the then-part is true.

Example (6)

If the Riemann Hypothesis is true, then $x^2 \geq 0$ for every real number x .

The answer is true (and it does not matter if Riemann Hypothesis is true or false).

What about the following example?

- **If** I am a vegetarian, **then** I won't have chicken tonight for dinner.
 - **True.** This logically follows.
- **If** 30,000,000 is a multiple of 375, **then** so is 60,000,000.
 - **True.** If $30,000,000 = 375k$, then $60,000,000 = 375(2k)$, so 60,000,000 would also then be a multiple of 375.

Truth table for

$$P \implies Q$$

P	Q	$P \implies Q$
T	T	T
T	F	F
F	T	T
F	F	T

- ▶ The implication "If I am a vegetarian, **then** I won't have chicken tonight for dinner" certainly "restricts" vegetarians having chicken for a dinner.
- ▶ However non-vegetarians (last two lines in the truth-table) can have either chicken or anything else for dinner. The statement will still be True.
- ▶ What if we wanted to restrict the option of "*not having a chicken*" for dinner to only vegetarians (i.e. *non-vegetarians must eat chicken regardless*)?
- ▶ In this case we must say "**Only if** I am a vegetarian, then I will not have chicken ..."
- ▶ Mathematicians make it more dramatic and instead of saying "**only if**" they say "**if and only if**"
- ▶ Notice that now if you are **not having chicken**, that means **you are certainly a vegetarian**. Hence both parts of this proposition (if part and then part) becomes equivalent!
- ▶ it becomes an implication both ways: "if you are vegetarian $\implies$ you didn't have chicken" and "if you didn't have chicken $\implies$ you are (certainly) a vegetarian".

The proposition " **P if and only if Q** " asserts that **P and Q are logically equivalent**; that is, either both are true or both are false. In other words:

$(P \implies Q) \wedge (Q \implies P)$ is equivalent to $P \longleftrightarrow Q$.

For some values of x , both inequalities are true. For other values of x , neither inequality is true. In every case, however, the proposition as a whole is true.

Example (7)

$$x^2 - 4 \geq 0 \longleftrightarrow |x| \geq 2$$

Truth table for "IFF":

P	Q	$P \longleftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Notations:

English	Symbolic
NOT(P)	$\neg P$ or $\overline{P}$
P AND Q	$P \wedge Q$
P OR Q	$P \vee Q$
P IMPLIES Q	$P \implies Q$
if P then Q	$P \implies Q$
P IFF Q	$P \longleftrightarrow Q$

Quantifiers:

$\exists$	There exists – Existential quantifier
$\forall$	For all – Universal quantifier

Note: $P \implies Q$ is logically equivalent to $\neg P \implies \neg Q$

► P **and** Q — true when both P and Q are true statements.

- **Example:** I am from the United Kingdom, and I studied maths.

► P **or** Q — true when at least one of P or Q is true — or both!

► **Example:** I am a doctor of philosophy, or I am a doctor of medicine.

► **If P , then Q** — The statement Q logically follows from P .

- **IMPORTANT:** *This does not mean that P itself is true!!!*

► **If and only if P then Q** — Q follows from P , and P follows from Q .

- ▶ Let's think back to our first example...

Theorem. 15 is a multiple of 3.

- ▶ To say that a (whole) number x is a multiple of 3 means that there is some other (whole) number n for which $3 \times n = x$.
- ▶ So the theorem can be rewritten as:

Theorem. There is a whole number n for which $3 \times n = 15$.

- ▶ This is an existential statement: it is a statement that asserts the *existence* of a number with a particular property (in this case, the property of the number is that its product with 3 is 15).

Example (8)

There exists ($\exists$) natural number n such that $n^2 = 169$. $\exists n \in \mathbb{N} : n^2 = 169$

- ▶ Is this statement true?
 - Yes! This works for $n = 13$, since $13^2 = 169$.

Example (9)

There are numbers a, b, c for which $a^2 + b^2 = c^2$.

- ▶ Is this statement true?
 - Yes! We can take $a = 3$, $b = 4$, $c = 5$, since $3^2 + 4^2 = 9 + 16 = 25 = 5^2$, so $3^2 + 4^2 = 5^2$.
 - This also works with $a = 5$, $b = 12$, and $c = 13$: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$.
- ▶ Existential statements do not necessarily say that the object whose existence is asserted is unique: there may be more than one in question (such as was the case in the last Example 11).

- ▶ **Question.** How do we prove that an existential statement like “There are numbers a, b, c for which $a^2 + b^2 = c^2$ is true?”
- ▶ **Answer.** Find (at least one) example! In the above, $a = 3, b = 4, c = 5$ does the trick, and shows that such an a, b , and c do exist.
- ▶ In general, to prove an existence statement, only one example needs to be produced.
- ▶ But in some cases, proving the existence may be very difficult...

Example

There exist numbers a, b, c for which $a^3 + b^3 = c^3$.

- ▶ Is this statement true? **NO!** ... Actually, it took mathematicians hundreds of years to answer...
- ▶ Another example:

👉 **Euler**

conjecture: $a^4 + b^4 + c^4 = d^4$ has no solution for positive integers a, b, c, d .

It took 218 years to find a solution satisfying this equation and prove the proposition is false! ($a = 95800, b = 217519, c = 414560, d = 422481$)

The true statement, it turns out, is:

Theorem

There are no numbers a, b, c for which $a^3 + b^3 = c^3$.

- ▶ But how do you prove it? You certainly can't check it with every possible triple of numbers a, b, c .
- ▶ You need to show that something is true for every choice of a, b, c : that $a^3 + b^3 \neq c^3$.
- ▶ So another way to write the statement is:

Theorem

For all numbers a, b, c : $a^3 + b^3 \neq c^3$.

- ▶ Statements that make claims about *all* numbers are **universal statements**.
- ▶ See the difference: An **existential statement** claims that a certain property holds for **some** number; perhaps one, perhaps several, perhaps all, but at least one. A **universal statement** claims that a certain property holds for **all** numbers.

- ▶ The statement “For all numbers a, b, c : $a^3 + b^3 \neq c^3$ is exceptionally hard to prove.
- ▶ Actually, it’s a specific case of a more general, very famous universal statement:

Theorem. For all numbers n , there are no numbers a, b, c for which $a^n + b^n = c^n$.

- ▶ Let’s look at a simpler example:

Example

For all whole numbers n , the number $n^2 + n$ is even.

- ▶ Is this true?
- ▶ It turns out the answer is yes:

Theorem. For all numbers n (i.e. $\forall n \in \mathbb{N}$), the number $n^2 + n$ is even.

Proof.

- ▶ We need to show that there is an integer k for which $n^2 + n = 2k$.
- ▶ A number n is either even or odd.
- ▶ Case 1: n is even, say $n = 2r$
 - Then $n^2 + n = 4r^2 + 2r = 2(2r^2 + r)$, which is even: $\therefore k = 2r^2 + r$.
- ▶ Case 2: n is odd, say $n = 2r + 1$.
 - Then $n^2 = (2r + 1)^2 = 4r^2 + 4r + 1$. So,

$$n^2 + n = \underbrace{4r^2 + 4r + 1}_{n^2} + \underbrace{2r + 1}_n = 4r^2 + 6r + 2 = 2(2r^2 + 3r + 1).$$

So again, $n^2 + n$ is even: $k = 2r^2 + 3r + 1$. □

- ▶ This is often a common way of proving universal statements: show that there are **finitely many** cases, and show that it works in each such case.

Proof by contradiction, also known as *reductio ad absurdum*, is a powerful indirect proof technique used to establish the truth of a statement.

► The Steps:

- 1 Assume the **negation** of the statement you want to prove is true.
- 2 Use a logical chain of reasoning to follow from this assumption.
- 3 Arrive at a **contradiction**. This is a statement that is logically impossible, such as $x > 0$ and $x \leq 0$.
- 4 Since a false premise must have led to this contradiction, our initial assumption must be false.
- 5 Therefore, the original statement we wanted to prove must be true.

► Example: Proving $\sqrt{2}$ is Irrational

- ① **Statement:** $\sqrt{2}$ is irrational.
- ② **Assume the opposite:** $\sqrt{2}$ is rational, i.e. $\sqrt{2}$ can be expressed as a rational fraction.
- ③ Then by definition, $\sqrt{2} = \frac{p}{q}$ where p and q are integers, $q \neq 0$, and $\frac{p}{q}$ is in its simplest form.
- ④ Squaring both sides: $2 = \frac{p^2}{q^2} \implies 2q^2 = p^2$. This means p^2 is even, so p must be even.
- ⑤ Let $p = 2k$ for some integer k . Substituting this gives $2q^2 = (2k)^2 = 4k^2 \implies q^2 = 2k^2$.
- ⑥ This means q^2 is even, so q must also be even (any product with even number is even).
- ⑦ **Contradiction!** We have shown that both p and q are even, which contradicts our initial assumption that the fraction $\frac{p}{q}$ was in its simplest form.
- ⑧ Therefore, the original statement is true: $\sqrt{2}$ is irrational.

- ▶ Let's review some of the basic proof techniques we've seen so far:
- ▶ **To prove an existential statement is true, produce an example.**
 - E.g., "15 is a multiple of 3" (one example is enough).
- ▶ **To prove a universal statement is false, produce a counterexample.**
 - E.g., "For all numbers n , $4n - 1$ is prime" was proven false because it fails to be true when $n = 4$.
- ▶ These can still be difficult to do in general.
- ▶ Another powerful technique is "**proof by contradiction**": assume the statement you're trying to prove is false, and arrive at a false conclusion.
 - E.g., "20 is not a multiple of 3" was proven by contradiction.
- ▶ This list is not exhaustive! There are many other types of proofs. Some are easier or harder than others. But there is no strategy that works for proving every true statement.

- ▶ We will be continuing with logic, rules of inference and proofs next week.
- ▶ Finish Quiz 1 ! (Quiz 2 on Logic will be given on next Tuesday)
- ▶ End of this week: Prepare Tutorial questions for your 1st Tutorial Session: Tutorial Sheets can be found on Aula in Week 1 section.
- ▶ One of the Tutorial groups will have their first Tutorial on Monday!